



Evidence Based Practice for Risk Management & Quality Improvement

By

Dr/Nehal Mahmoud

Introduction

- Evidence-based practice (EBP) is the process of **collecting, processing, and implementing best research findings to improve clinical practice, the work environment, or patient outcomes.**
- According to the American Nurses Association (ANA), **nursing interventions should be practical, methodical decisions based on EBP research studies.** Utilizing the EBP approach to nursing practice helps us provide the highest quality and most cost-efficient patient care possible.

Definition of Evidence Based Practice:

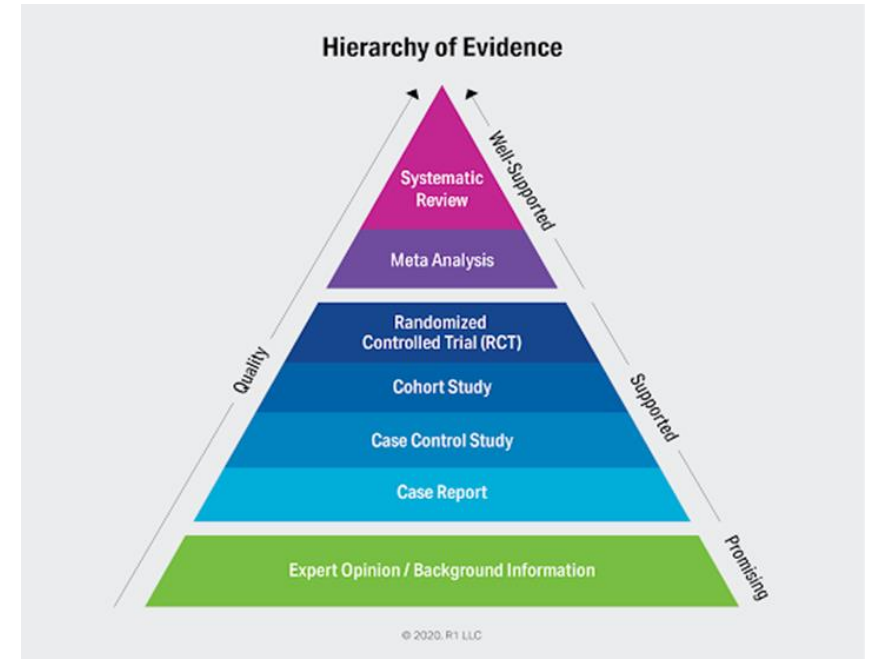
Evidence-based practice is a systematic approach that emphasizes the use of **best evidence, clinical experience** and **patient preferences** to make decisions about patient care and treatment that leads to the best patient outcomes, by selecting interventions that have the greatest chance of success.

EBP is a process used to review, analyze and translate the latest scientific evidence.

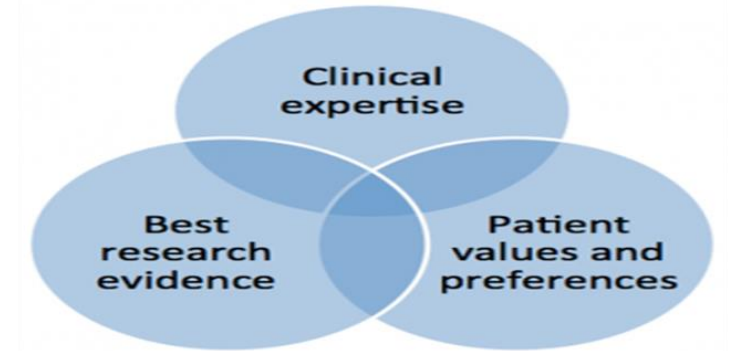
EBP has **one** goal, **two** fundamental principles, **three** components and **five** steps.

- **One goal** is to improve the quality of clinical care through providing the most effective and efficient care possible. It also **helps to reduce variation in care** and **improve patient outcomes**.

Two principles are hierarchy of evidence and insufficiency of evidence alone in decision-making.



- **Three components** are evidence, expertise and expectations of patients (triple **Es**).



Five steps are
asked, acquire,
appraise, apply
and assess (5
As).



The 3 main Components of Evidence Based Practice

- **Best Evidence:** evidence is simply **information** that supports a **conclusion**; in healthcare, this typically refers to the **effectiveness of interventions** in achieving specific outcomes.
- **Clinical Experience:** It's the ability to **use our clinical skills and experience** to rapidly identify each patient's unique health state and **diagnosis, their individual risks and benefits of potential interventions** and their personal values and expectations.
- **Patient Preferences:** the unique **preference, concerns and expectations each patient** brings to a clinical encounter, and which **must be integrated into clinical decisions** if they are to serve the patient.

Three Main Components of Evidence-Based Practice



Utilize the best
external evidence



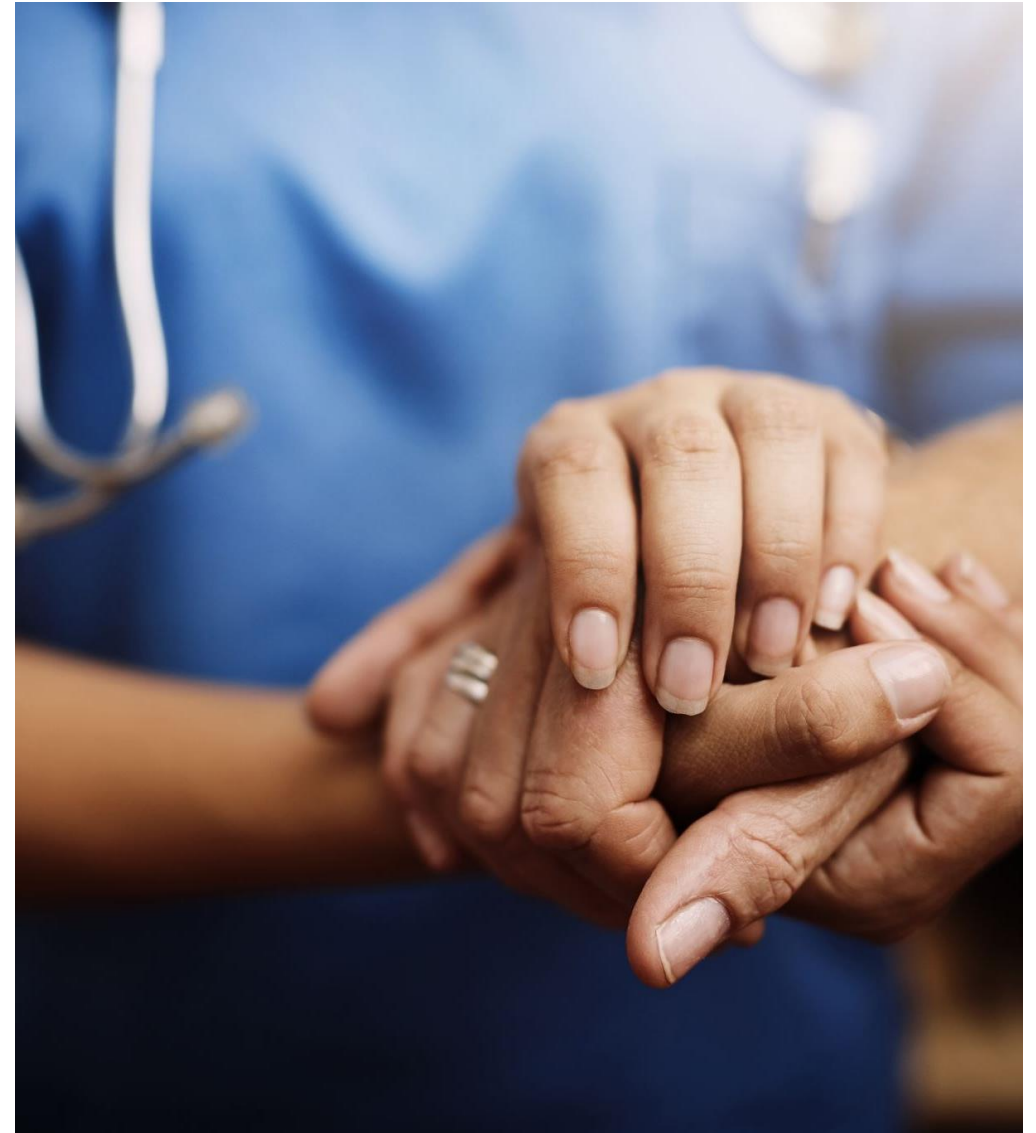
Draw on individual
clinical expertise



Consider patient
values & expectations

Benefits of Evidence Based Practice:

- Evidence-Based Practice has many benefits for **patients, nurses, and health care organizations.**
- **1- For patients:**
- Improved care and reduced time wasted on inappropriate treatments
- Enhances consistency in care
- Increases understanding of investigations and treatment
- Increases accountability for care provision



2- For clinical Practitioners / nurse:

- Actively involves them in determining the appropriateness and effectiveness of care
- Help in redefining and/or changing practice where necessary
- Enhances quality of care
- Presents best evidence of practice to patients and carers
- Increases confidence in practitioners



3- For the health care organization

- Enables consistent decision making
- Reduces variation in services
- Promotes cost-effectiveness
- Promotes integration of activities (e.g. research and development, clinical audit, continuing professional development)
- Increase accountability to the public for the service provided

Five Key Steps of Evidence Based Practice (5As)

Step (1) Develop a researchable question (Ask):



Asking questions in a particular format is an important first step. It specifies the outcomes you are looking for. It also helps in assessing the relevance of the information to your patient's problem.



The **PICO** model is often used to define a problem and formulate a specific question: **P**opulation; **I**ntervention; **C**omparison; **O**utcome.

An example of the PICO is as follows:

In hospitalized patients over **60 years** of age, how effective is **a falls prevention** program in comparison to the **normal standard** of care in **decreasing falls and falls injury** rates?



P (Population): Hospitalized patients over 60 years of age

I (Intervention): Falls prevention program

C (Comparison): Standard (usual) care

O (Outcome): Reduction in falls and fall-related injury rates

PICO Question:

In hospitalized patients over 60 years of age, how effective is a falls prevention program compared to standard care in reducing falls and fall-related injuries?

Step (2) Select resources and conduct the search (Acquire):

- In this stage **searching for evidence.**
- Information is **gathered from several sources** including: systematic reviews, clinical practice guidelines and searching journal publications for pertinent research articles.

Sources of evidence	Actions
Research	Search published and unpublished literature – original research (primary sources), reviews (secondary sources), clinical guidelines (e.g. NICE), published research-based standards of care Generate own evidence through research
Experience (professional or general)	Reflect on practice: articulate reflections, facilitate discussion, search the literature
Theoretical experience (not research-based)	Search the literature (published or unpublished), learn from others, facilitate discussions
Evidence gathered from patients and/or their carers	Search the literature for experiential writings and/or research findings, use audit data, levels of satisfaction and/or complaint, facilitate discussion and collaborative decision making
Evidence passed on from role models and/or experts	Facilitate discussion and observation, consult with experts. Search the literature for findings from consensus-reaching techniques (focus groups, Delphi surveys, nominal groups technique and consensus conferences)
Evidence derived from policy documents	Scrutinize documentation, facilitate discussion

Step (3) Review and appraise evidence (Appraise):

- **Assess the strength of the evidence.** We need to know which form of evidence is '**best**' and how we reach conclusions about its quality.
- **Critical Appraisal** is the way to evaluate any research, **to be sure that:**
- **Validity:** The evidence is telling the truth.
- **Relevant:** The evidence should directly answer the question
- **Applicability:** Can the results be applied to local population.



Step (4) Applying the evidence (apply):

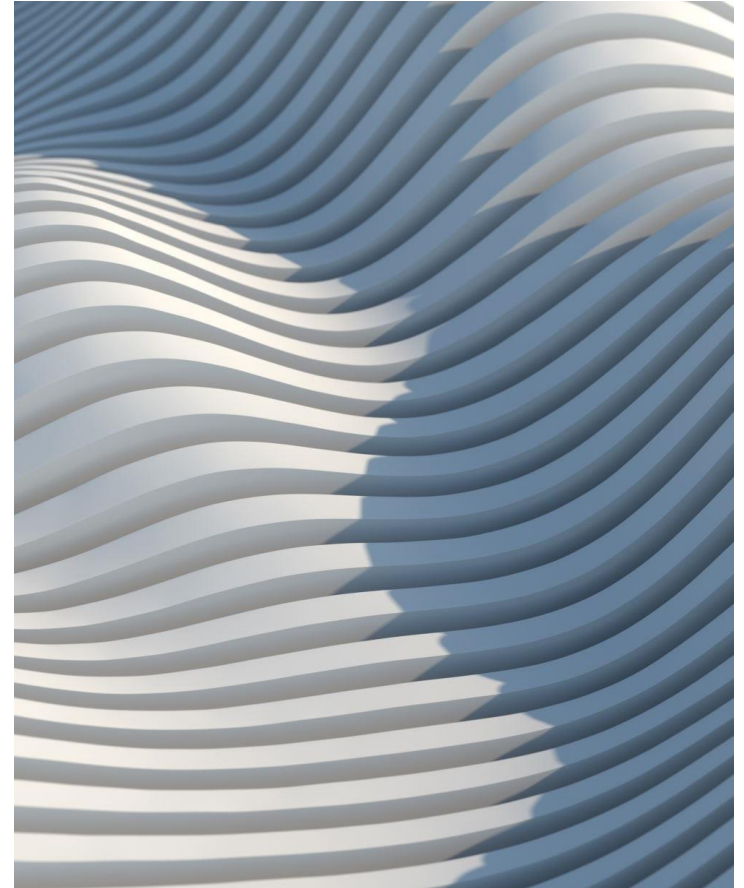
choose interventions and select the most valid evidence

search and integrate the evidence with one's clinical expertise and the unique needs and values of target patient through:

GUIDLINE: that collaboratively communicates regular reviews & updates

PATHWAY: this describes how patient moves through his treatment

PROTOCOLS: instructions for practitioners to safeguard patients as any deviation from it needs explanation and policies



Step (5) (Assess) Evaluate the outcome of evidence-based practice changes:



Determine how the practice change is used. Audit and feedback demonstrate the gap between actual and desired results and address questions such as:



Did the clinicians receive the information about a practice change?



Did they adhere to the practice change?



How difficult or smooth was it to use the new way.

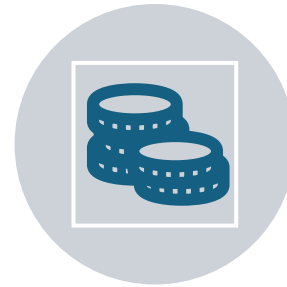


In this stage, evaluate the impact of EBP on patients, efficacy, efficiency, economic analysis, & health status impact.

When evaluating the impact, consider the following:



The client's outcomes.



Cost effectiveness.



Client satisfaction.



Health care providers satisfaction.

Common Challenges in Applying Evidence-Based Practices



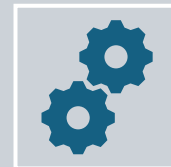
1. Limited access to evidence-based guidelines and research



2. Insufficient time or resources to implement evidence-based practices



3. Resistance to change: Healthcare providers may be hesitant to adopt new practices



4. Patient factors: Patient preferences, values, or circumstances may influence the application of evidence-based practices

Strategies to Overcome Challenges

1

1. Provide education and training:

Educate healthcare providers on evidence-based practices and guidelines

2

2. Develop clinical decision support tools:

Create tools to guide clinical decision-making

3

3. Implement policies and procedures:

Develop policies and procedures to support evidence-based practices

4

4. Encourage interdisciplinary collaboration:

Foster collaboration among healthcare professionals to promote evidence-based practices